Project Report

I Revolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau

Team Members

Team ID: LTVIP2026TMIDS36490

Team	Names
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1. INTRODUCTION

1 Project Overview

I Revolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau" is an analytical project that studies the growth and influence of the Apple Inc. iPhone in the India market. Using Tableau, the project transforms sales data, demographic information, and social media insights into interactive visualizations. It highlights market penetration, sales performance, customer preferences, and cultural influence of the iPhone across different regions of India.

Purpose

The main purpose of this project is to analyse how the iPhone has impacted the Indian smartphone market and consumer behavior. It helps stakeholders understand sales trends, identify key customer segments, and evaluate brand perception. The insights support better decision-making for marketing strategies, competitive positioning, and future business growth.

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love A well-articulated customer problem statement allows you and your team to find the ideal

solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	A value-conscious, aspiaronal Indian consumer
I'm trying to	Own an premium smartphone that offers long-term value,high-quality features, and social status
but	High price barriers and taxes make signficantly expensive in India comparied other markets
because	I deseire yo be part to the exclusive Apple ecosymem and gain social oin social recognition without severe financial strain
which makes me feel	Anxious about the cost, yet highly motivated and aspizraral to mpact them purchase

example

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A tech-savvy Indian consumer	Buy a premium iPhone for performance, ecosystem, and status	High prices and taxes make iPhones expensive in India	Import duties and premium branding increase overall cost	Anxious about affordability but still aspirational
PS-2	A college student / young professional	Own an iPhone to access Apple ecosystem and better features	Limited budget and EMI depende **ncy**	iPhones are positioned as luxury products Financial pressure yet motivated to purchase	Financial pressure yet motivated to purchase

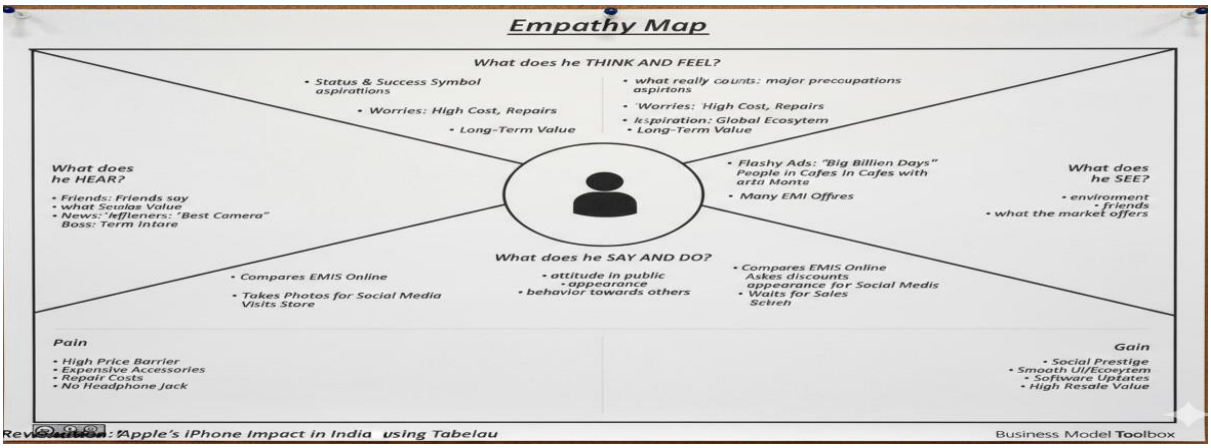
2.Empathy Map Canvas

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

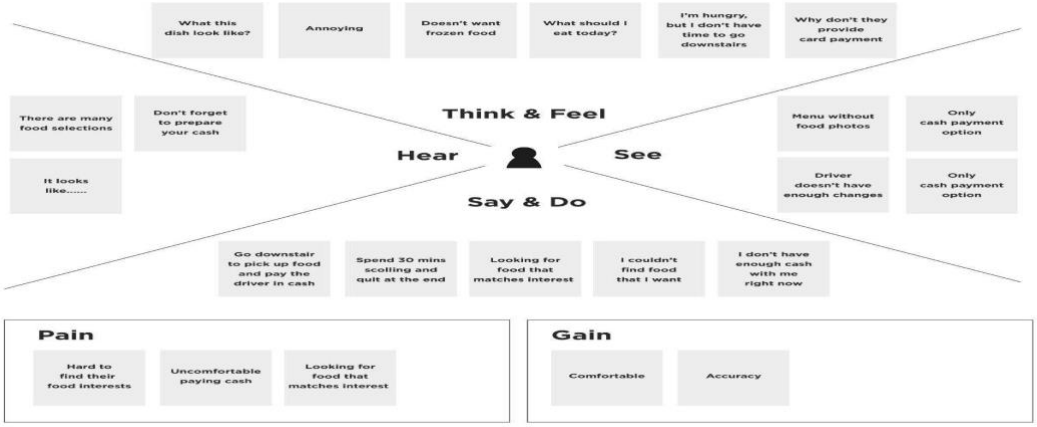
It is a useful tool to helps teams better understand their users

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Example: I revolution: a data-driven exploration of apple's iPhone impact in India using



3 Brainstorming Brainstorm & Idea Prioritization

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room. His helped us focus on the most valuable insights for building the dashboard.

Then, we grouped similar ideas to identify key features for the dashboard like skin type filters, price analysis, and top brand insights.

step:2 Brainstorm, Idea Listing and Grouping

Step-3: Idea Prioritization

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

URL for customer journey map

3.2 solution Requirements

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	1. Collect iPhone sales data in India 2. Collect revenue & pricing data 3. Collect market share data
FR-2	Data preparation	1. Clean missing values 2. Remove duplicate records 3. Format data for Tableau
FR-3	Dashboard development	1. Create sales trend charts 2. Create market share comparison charts 3. Create revenue growth visualization 4. Add filters (Year, Model, Region)
FR-4	Data analysis	1. Identify growth patterns 2. Compare iPhone with other brands 3. Analyse pricing strategy impact

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be simple, clean, and easy to understand
NFR-2	Security	Data should be stored securely and protected from unauthorized access
NFR-3	Reliability	Dashboard should display accurate and consistent results

NFR-4	Performance	Dashboard should load within few seconds
NFR-5	Availability	Project file should be accessible anytime for presentation
NFR-6	Scalability	System should handle additional future data (new iPhone models, new years)

Problem–Solution Fit

3. TRIGGERS TR

Rapid growth of iPhone sales in India News about Apple manufacturing expansion in India

Reports on increasing premium smartphone demand

4. EMOTIONS: BEFORE / AFTER EM

How do customers feel when they face a problem or a job and afterwards?

Confused about large raw datasets

Overwhelmed by unorganized sales data

Unsure how to identify meaningful trends

Lack of clarity about iPhone's real impact in India

Nervous about presenting insights to faculty

Worried about time-consuming data cleaning

10. YOUR SOLUTION

Large amounts of raw iPhone sales data are difficult to interpret

No clear visualization of growth trends in India

Hard to understand revenue, market share, and regional performance

Students lack structured insights for academic presentation

8. CHANNELS of BEHAVIOUR CH

1. ONLINE

- Use Tableau software Upload and analyze datasets Create dashboards Share dashboards digitally (PDF / PPT / Screenshot)

2. OFFLINE

Download datasets from online sources Read market research reports Refer to Apple India sales statistics Access academic journals & articles

Purpose:

"I Revolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau" is to transform complex and unstructured iPhone sales data into clear, meaningful, and interactive visual insights. It helps students understand growth trends, revenue patterns, market share, and regional performance of Apple Inc. and its iPhone in India. By using Tableau dashboards, the project reduces confusion, improves clarity, and supports confident academic presentations with structured, data-driven analysis.

Proposed Solution

S no	Parameter	Description
1	Problem Statement (Problem to be solved)	Large volumes of raw iPhone sales and revenue data in India are difficult to interpret. There is no clear visualization to understand growth trends, regional performance, and market share. Students need a structured way to analyze and present this data effectively.
2	Idea / Solution description	Develop an interactive Tableau dashboard system that collects, cleans, processes, and visualizes iPhone sales data in India. The solution transforms raw datasets into meaningful charts, KPIs, and trend analysis for easy understanding and presentation.
3	Novelty / Uniqueness	The project combines real-world Indian market data with interactive Tableau dashboards. It focuses specifically

		on Apple's iPhone growth in India and provides comparative, filter-based, and insight-driven visual analytics.
4	Social Impact / Customer Satisfaction	Satisfaction Helps students and analysts clearly understand smartphone market trends. Improves data literacy and decision-making skills. Provides clear visual insights that enhance confidence during presentations.
5	Business Model (Revenue Model)	As an academic project, direct revenue is not generated. However, the solution can be extended into a data analytics consulting dashboard model for businesses to analyse smartphone market trends and consumer behaviour.
6	Scalability of the Solution	The solution can be scaled by adding more datasets (other smartphone brands, future years, regional expansion). It can be upgraded with predictive analytics and real-time data integration

.3 Solution Architecture

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chabot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudland etc.

7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local File system
8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Open source framework
	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint - 1	Data Collection	USN-1	As an analyst, I can collect the iPhone dataset from Excel/Google Sheets for analysis.	2	High
Sprint - 1	Data Preparation	USN-2	As an analyst, I can clean and validate the dataset before visualization.	3	High
Sprint - 1	Calculated Fields	USN-3	As an analyst, I can create calculated fields like Growth %, Market Share %, Revenue Change	3	High
Sprint - 2	Revenue Visualization	USN-4	As a user, I can view annual revenue trends in a line chart.	2	High
Sprint - 2	Market Share Analysis	USN-5	As a user, I can view quarterly and model-wise market share distribution.	3	High

Sprint - 2	Brand Comparison	USN-6	As a user, I can compare brand price and battery type using tree map and bar charts	2	Medium
Sprint - 3	Dashboard Design	USN-7	As a user, I can interact with filters (Year, Country, Model) on the dashboard.	3	High
Sprint - 3	Story Creation	USN-8	As a user, I can view insights in a step-by-step story format.	2	Medium
Sprint - 4	Web Integration	USN-9	As a user, I can access the dashboard through a Flask web application.	3	High
Sprint - 4	Performance Testing	USN-10	As a user, I can experience smooth dashboard performance without lag.	2	High

Total Story Points:

Sprint-1: 8
Sprint-2: 16
Total: 24

Project Tracker, Velocity & Burn down Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	5 Days	10 January 2026	15 January 2026	8	15 January 2026
Sprint-2	16	5 Days	16 January 2026	20 January 2026	16	20 January 2026

Velocity:

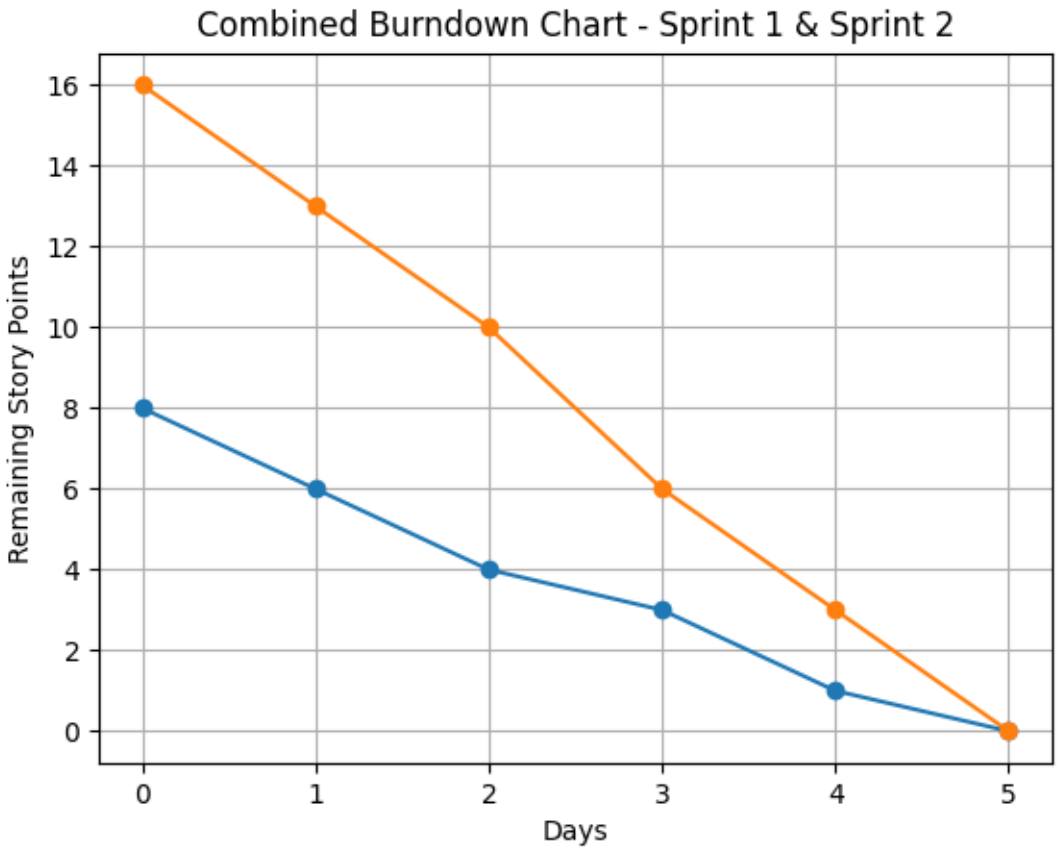
Total Story Points = 24

Number of Sprints = 2
Velocity = 24 / 2 = 12 Story Points per Sprint

Average Velocity per Day:

Sprint Duration = 5 Days
Velocity per Day = 12 / 5 = 2.4 Story Points per Day

Burn down Chart:



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	iPhone sales data, revenue trends, market share, regional performance charts successfully displayed in Tableau dashboards
2.	Data Preprocessing	Removed null values, formatted date fields, created calculated fields, cleaned duplicate records
3.	Utilization of Filters	Year filter, Region filter, iPhone Model filter, Price Category filter
4.	Calculation fields Used	Growth Rate (%), Total Revenue, Market Share %, Year-over-Year Growth
5.	Dashboard design	3 Dashboards – Sales Analysis, Market Share Analysis, Regional Performance (No. of Visualizations: 8–10 charts)
6	Story Design	1 Story with 5 Story Points explaining trends, insights, and conclusions

Key Performance Metrics

Metric	Description
Dashboard Load Time	Time taken for the dashboard to load completely after initial access
Visualization Rendering Time	Time taken to load maps, bar charts, and trend lines
Filter Response Time	Time taken to update visuals after applying filters (Region/Country/Category)
Calculated Fields Evaluation	Time spent computing growth rates, density metrics, and rankings
Data Volume	Number of UNESCO heritage records processed per worksheet

Test Results Summary

Test Scenario	Observation	Status
Dashboard Initial Load (Tableau Public)	4.5seconds on average	✅ Pass
Filter Response (e.g., Gender = Female)	1.2 seconds	✅ Pass
Story Scene Switch Time	2.1 seconds between transitions	✅ Pass
Visual Rendering with All Filters Applied	Slight lag on mobile, smooth on desktop	⚠️ Acceptable
Load on Flask Web Page	Fully rendered within 5–6 seconds (including embedded script)	✅ Pass

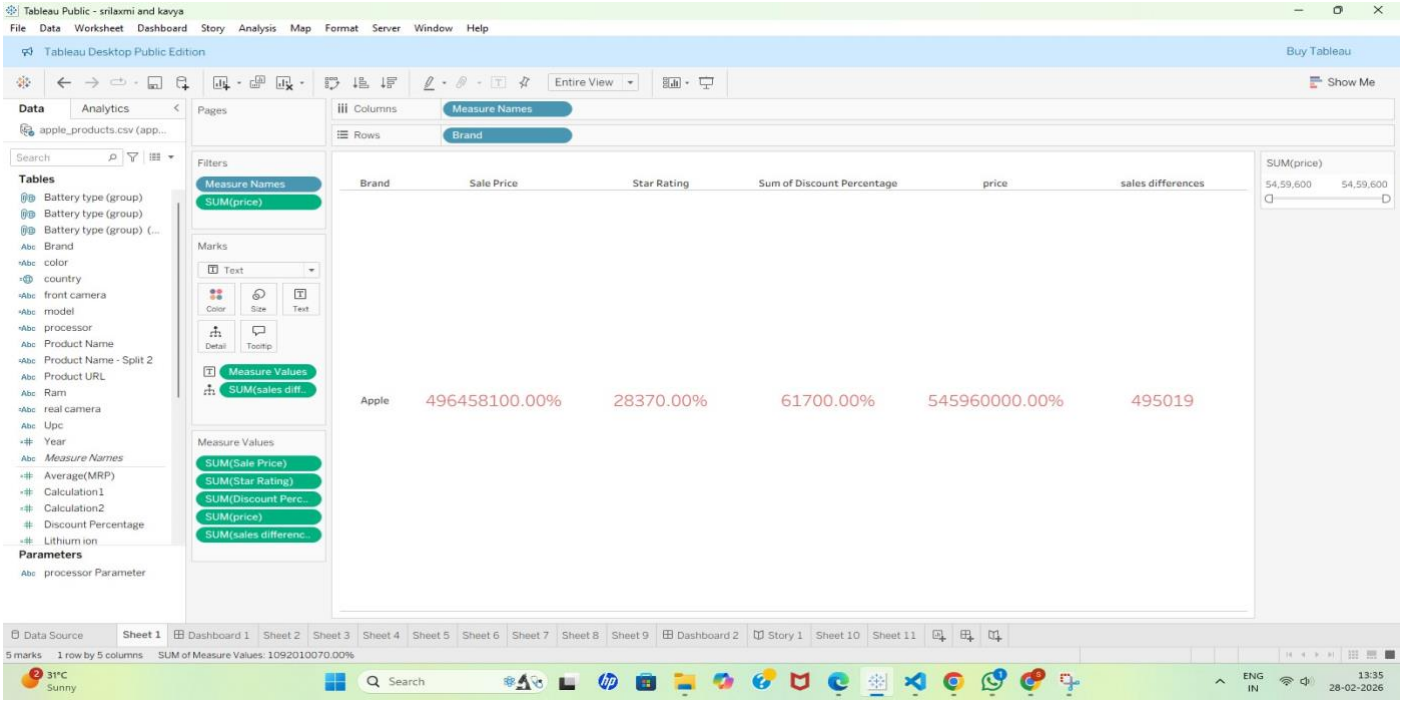
Recommendations for Optimization

Area	Optimization
Calculated Fields	Reduce complex LOD expressions and nested IF conditions
Filter Usage	Use Extract Filters instead of Context Filters when possible
Dashboard Layout	Avoid overcrowding dashboards with too many heavy map layers
Data Volume Handling	Pre-aggregate country-wise and region-wise totals before visualization

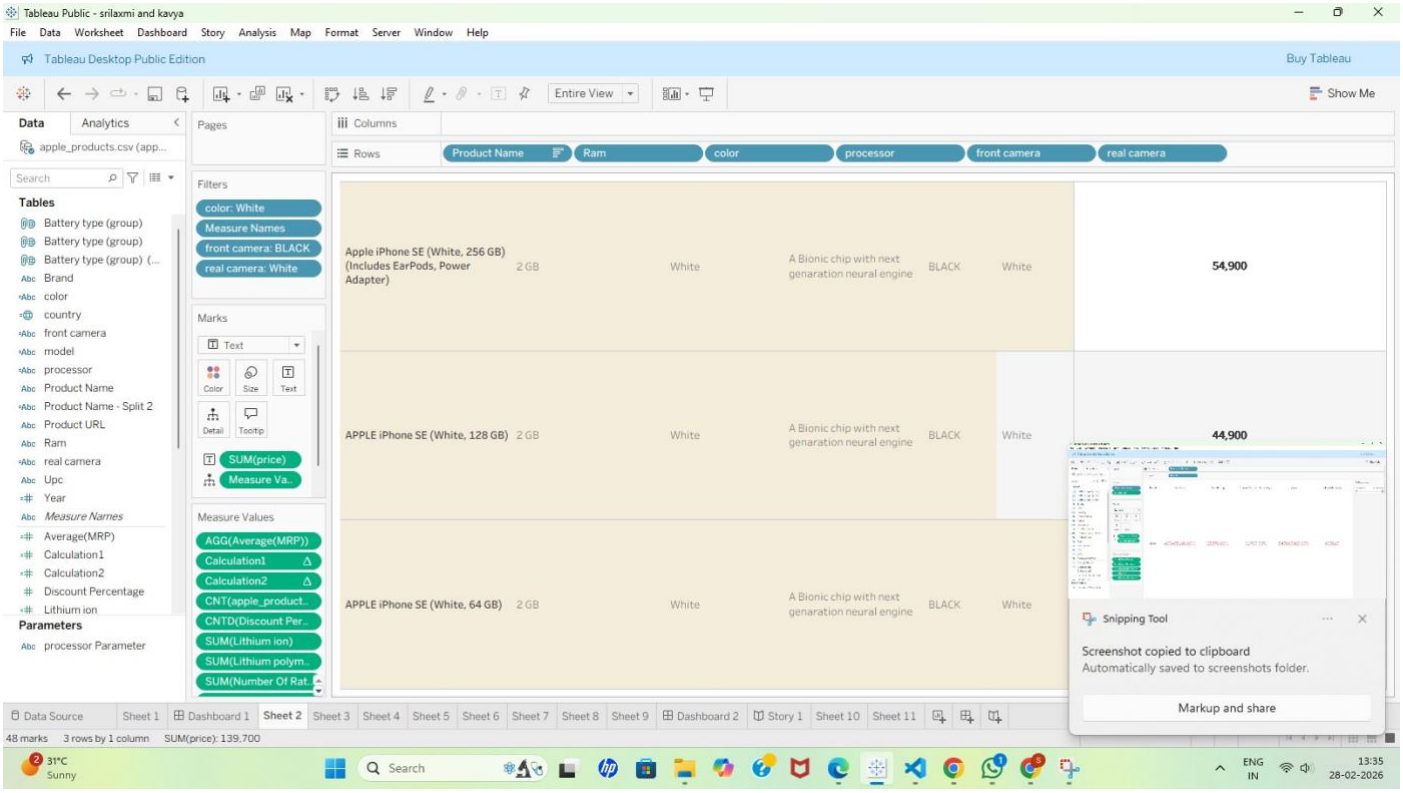
7. RESULTS

7.1 Output Screenshots

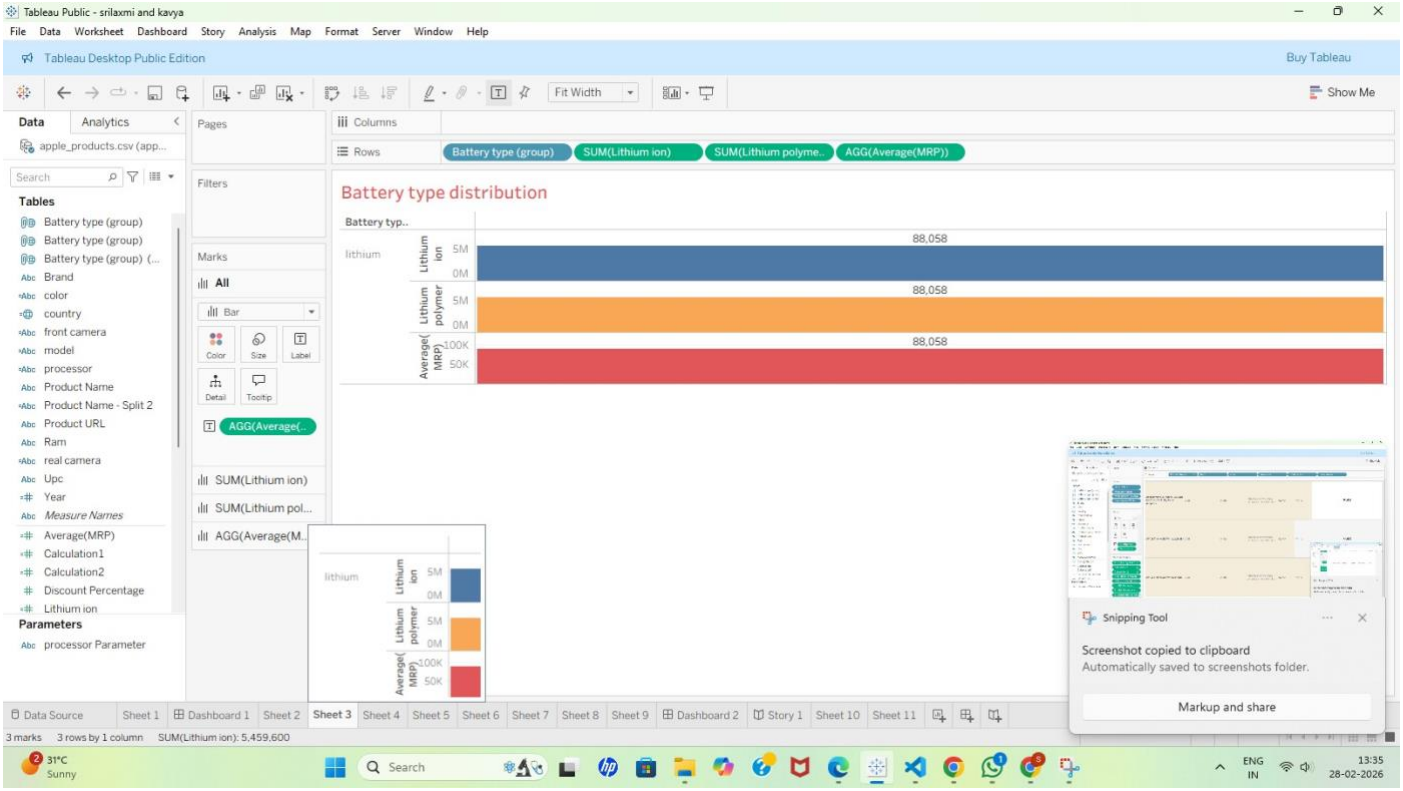
Activity 1.1: Countries per Region



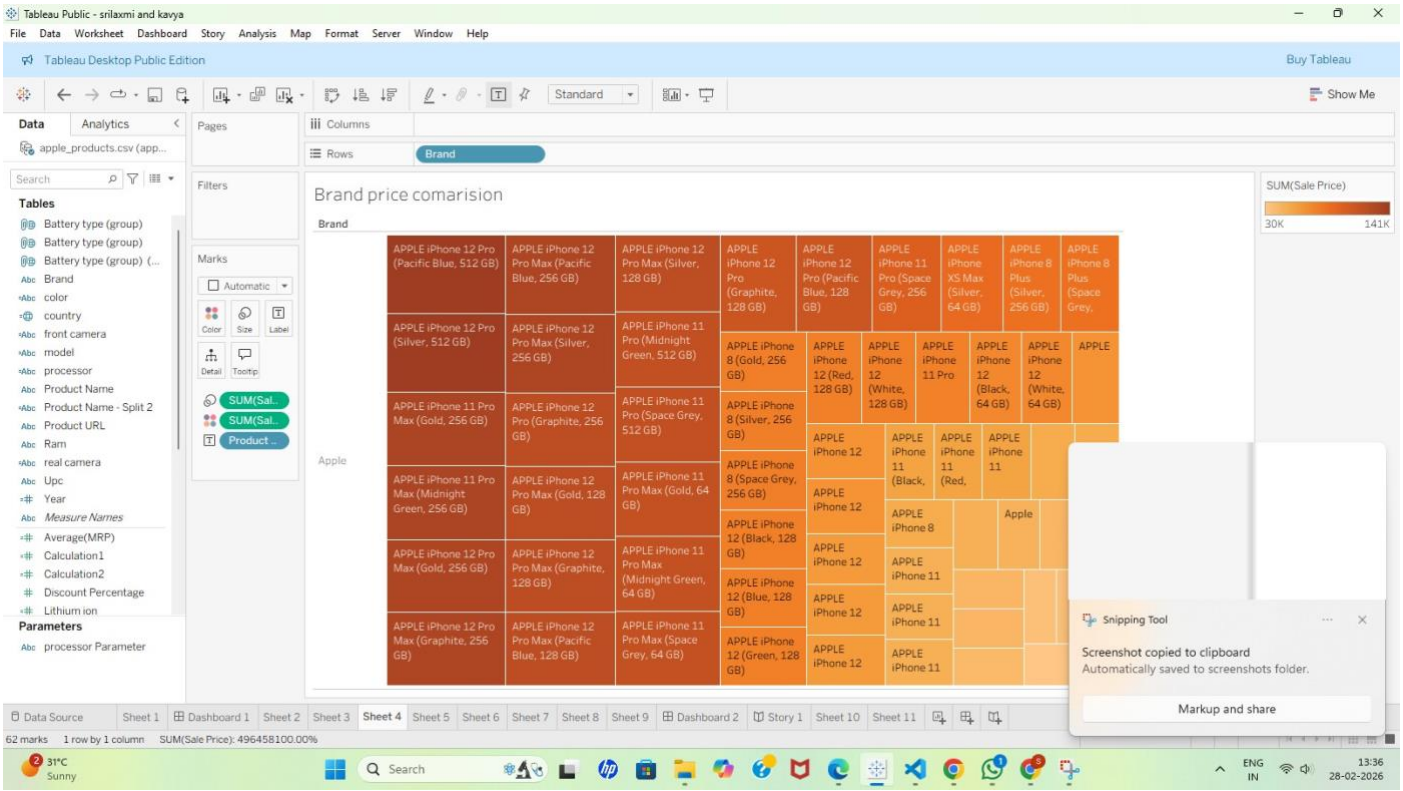
Activity 1.2: Top 10 Countries by Heritage Area



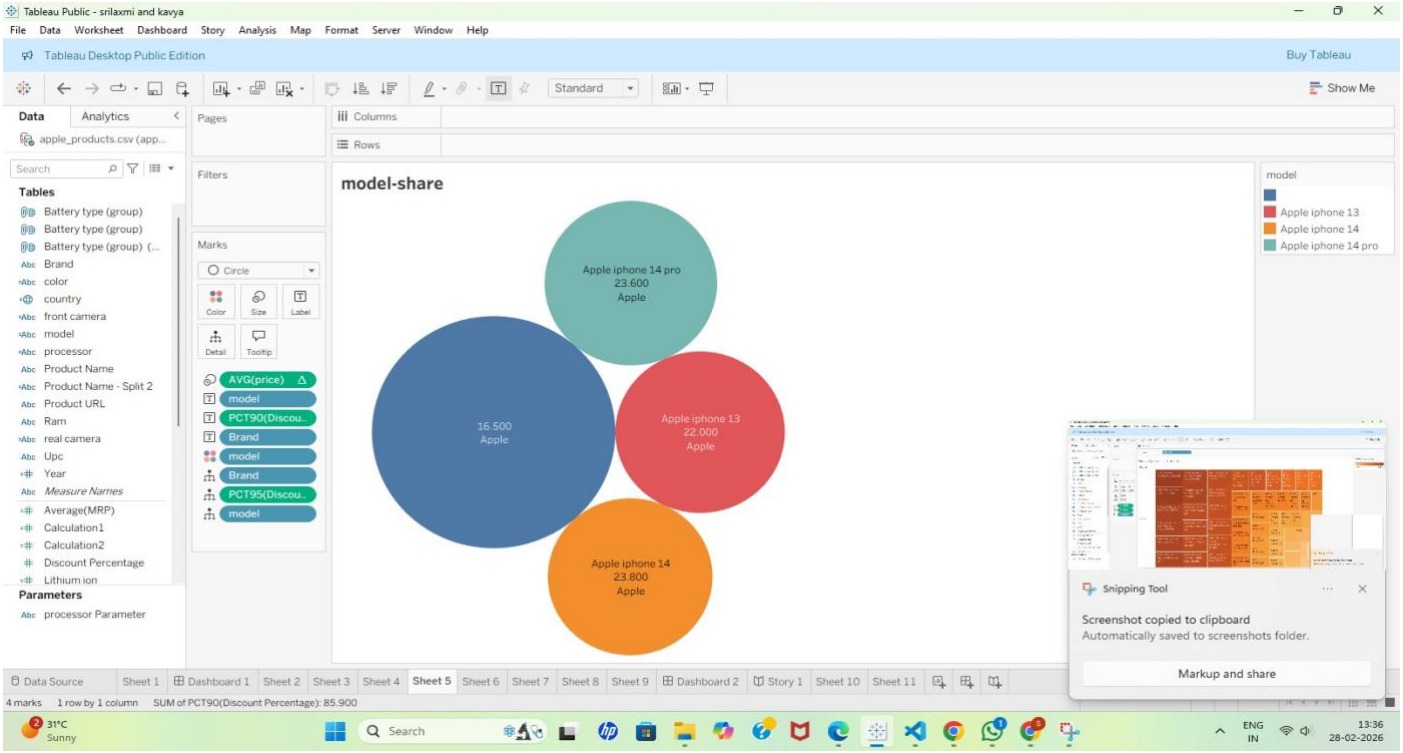
Activity 1.3: Region with Endangered Heritage Sites



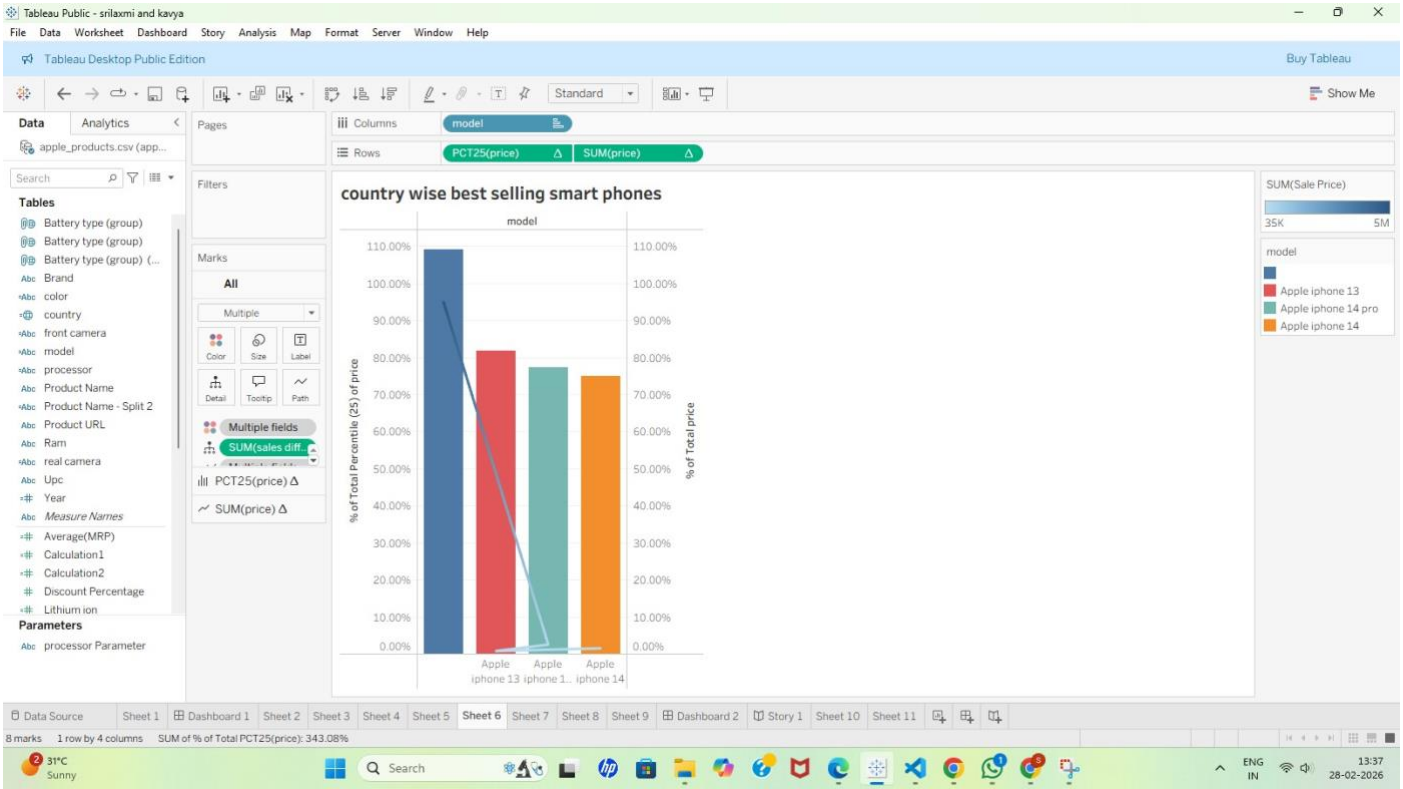
Activity 1.4 : Top 10 Danger Sites



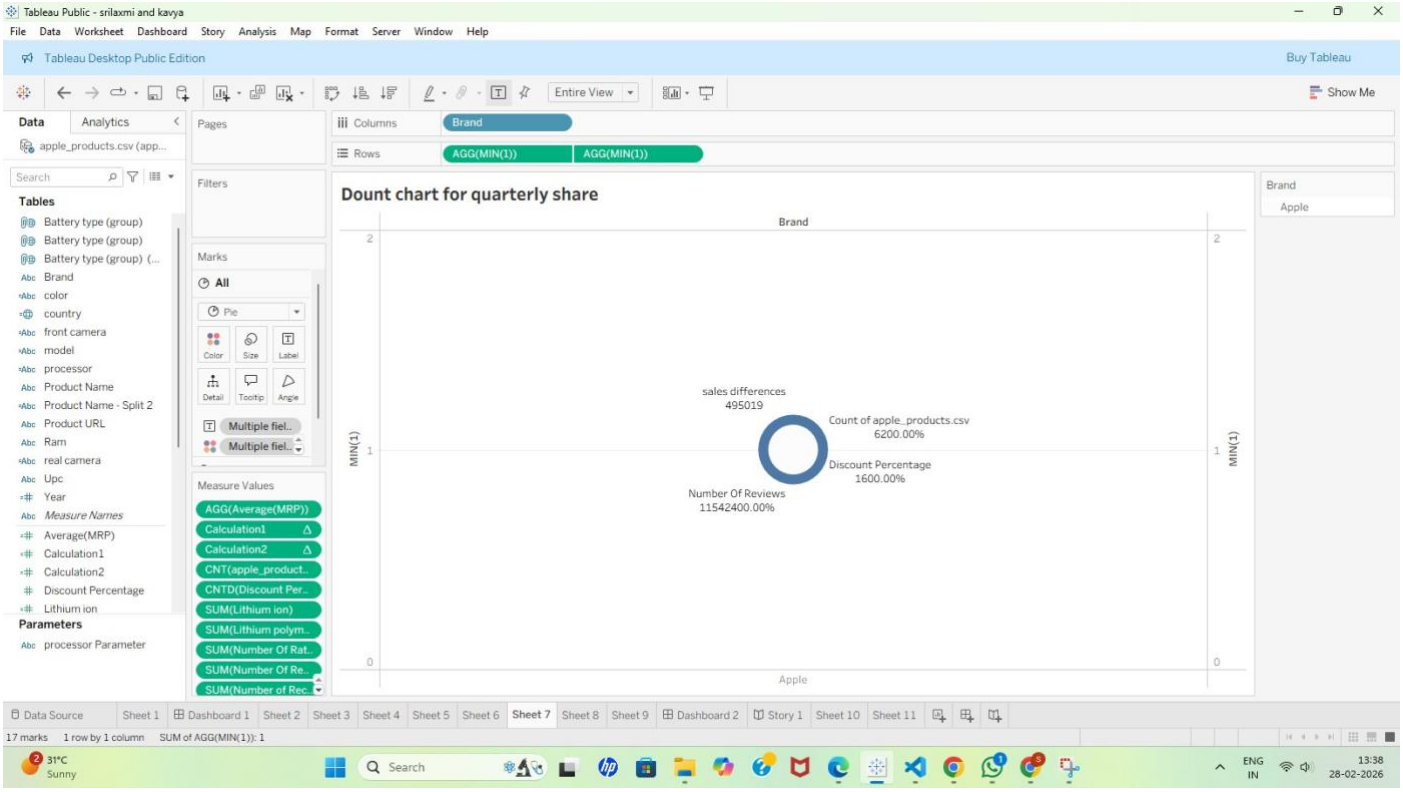
Activity 1.5 : World Heritage Site Forecast



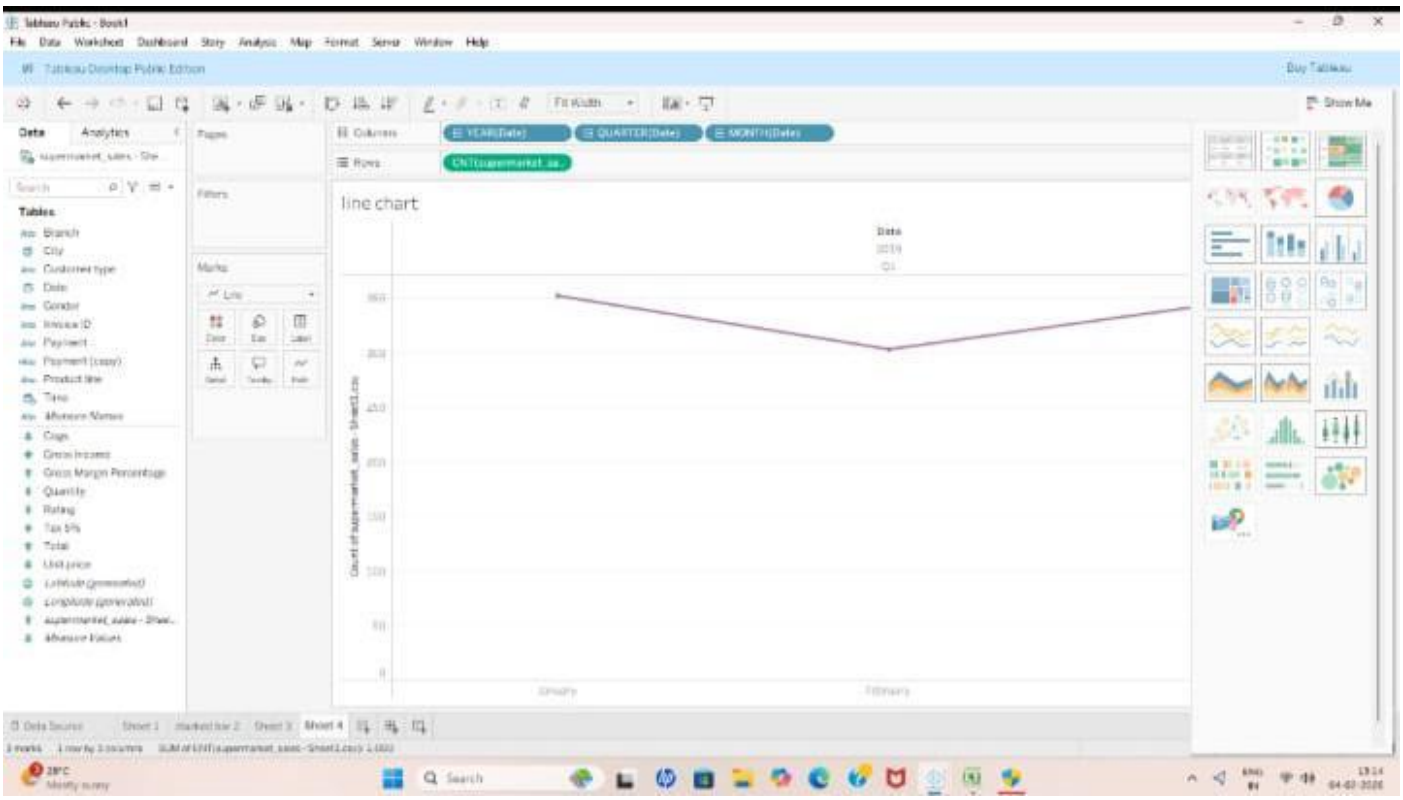
Activity 1.6 : Categories by Site Count



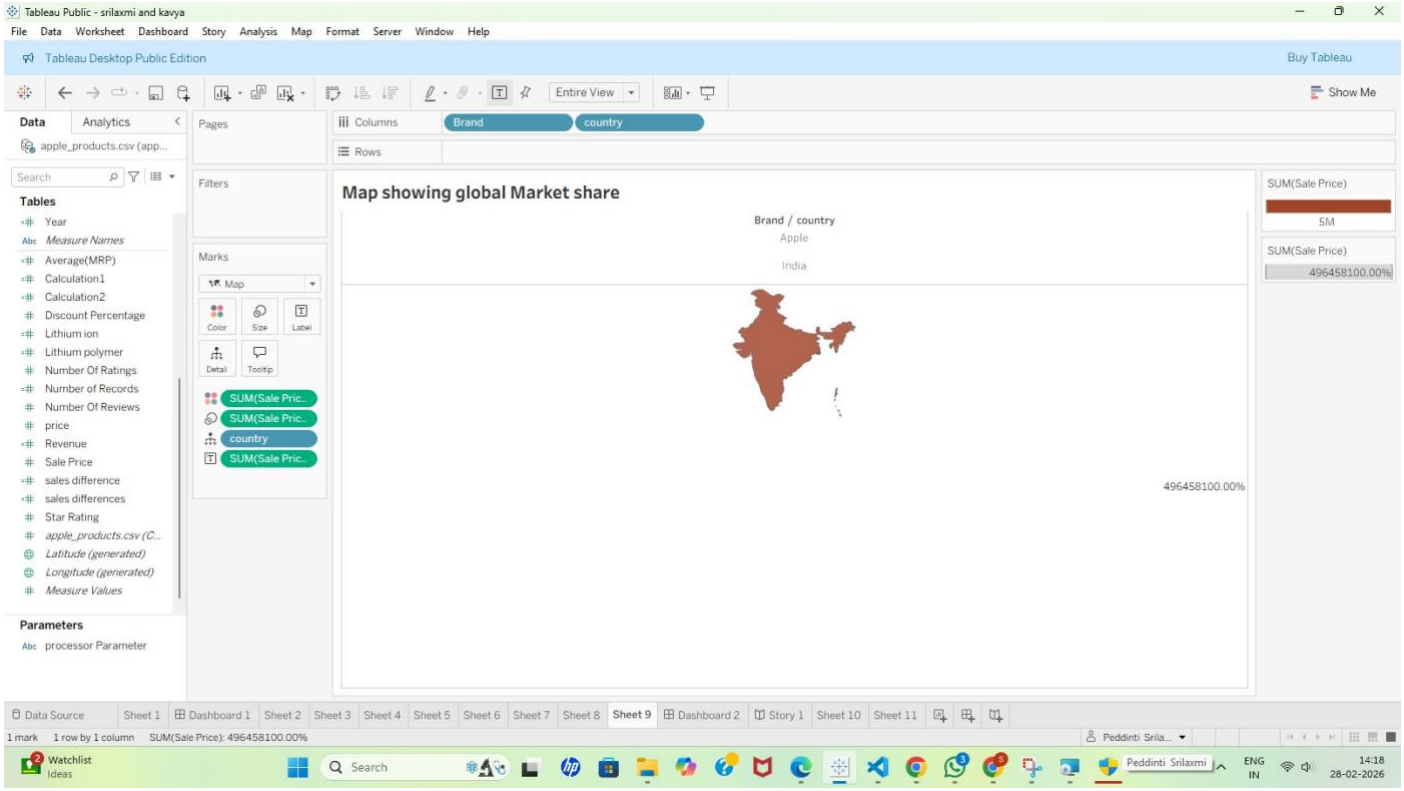
Activity 1.7 : UNESCO Site Count per Region



Activity 1.8 : Categories -Wise Area Distribution



Activity 1.9 : UNESCO World Heritage Explorer



Responsive and Design of Dashboard :

8. ADVANTAGES & DISADVANTAGES

Here is a detailed breakdown of the advantages and disadvantages of using Data Analytics with Tableau

Advantage	Description
1. Interactive Visualization	Using Tableau enables dynamic dashboards that make complex heritage data from UNESCO easy to explore and understand.
2. Data-Driven Insights	The project supports informed decision-making by identifying trends, regional distributions, and growth patterns of World Heritage Sites.
3. User-Friendly Interface	Intuitive design and interactive filters allow users from non-technical backgrounds to analyse data efficiently.

4. Scalability	**The data model can be easily updated with new heritage site inscriptions without major structural changes.**
5.Comprehensive Global Overview	**Provides a consolidated analysis of World Heritage Sites recognized by UNESCO in a single interactive platform.**
6. Advanced Data Visualization	**Utilizes powerful features of Tableau such as maps, trend lines, filters, and calculated fields for deeper insights.**
7. Improved Analytical Efficiency	**Reduces manual data analysis time by automating calculations and visual representations.**
8. Enhanced Comparative Analysis	**Enables easy comparison between countries, continents, and heritage categories (Cultural, Natural, Mixed).**
9. Better Strategic Planning	**Helps policymakers and tourism authorities identify growth opportunities and underrepresented regions.**
10. Customizable Dashboards	**Users can personalize views using filters and parameters according to their analytical needs.**

Disadvantages of with Tableau

Disadvantage	Description
1. Data Dependency	The accuracy of insights depends entirely on the completeness and correctness of the available UNESCO dataset.
2. Performance Limitations	Large datasets may affect dashboard performance if not optimized properly.
3. Limited Real-Time Updates	If static datasets are used, the analysis may not reflect the most recent heritage site additions.
4. Software Dependency	The project relies heavily on Tableau, which may require licensing costs and technical familiarity.
5. Data Quality Constraints	Incomplete or outdated data may lead to misleading interpretations.
6. Technical Learning Curve	Users may require basic training to fully utilize advanced Tableau features.
7. Limited Predictive Capabilities	The project focuses mainly on descriptive analytics rather than advanced forecasting models.

CONCLUSION

Project Conclusion

I Revolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau

This project successfully analyzed the impact of Apple's iPhone in the Indian smartphone market using Tableau data visualizations. By examining sales trends, revenue growth, market share, and regional performance, the project clearly demonstrated the rapid growth of iPhone adoption in India, especially in the premium smartphone segment.

The dashboards and story points helped transform complex raw data into meaningful insights, making it easier to understand growth patterns, consumer preferences, and year-over-year performance. The analysis shows that increasing demand for premium devices, expanding manufacturing in India, and strong brand value have significantly contributed to Apple's growth.

Overall, this project proves that data visualization tools like Tableau play a crucial role in converting large datasets into actionable business insights, supporting better strategic decision-making in the competitive Indian smartphone market.

You said:

ChatGPT said:

	10 February 2026
	PNT2022
	iRevolution: A Data-driven Exploration of Apple's iPhone Impact in India using Tableau
	_____

1	Data Rendered	iPhone sales data, revenue trends, market share, regional performance charts successfully displayed in Tableau dashboards
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5	Dashboard Design	3 Dashboards – Sales Analysis, Market Share Analysis, Regional Performance (No. of Visualizations: 8–10 charts)

6	Story Design	1 Story with 5 Story Points explaining trends, insights, and conclusions

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You said:

ChatGPT said:

The future scope of the Heritage Treasures project highlights its potential to evolve into a more intelligent, interactive, and sustainable platform for heritage management and tourism planning. By integrating AI, predictive analytics, IoT monitoring, and mobile accessibility, the system can provide real-time insights, personalized recommendations, and proactive conservation strategies.

These advancements will not only improve operational efficiency and stakeholder decision-making but also enhance visitor engagement and global collaboration for heritage preservation. With continued innovation, the project can serve as a comprehensive, scalable, and forward-looking tool to protect, promote, and analyze UNESCO World Heritage Sites worldwide.

11. Deployment of Flask Web Application with Embedded Tableau Dashboard

11.1 Overview

This section describes the deployment process of the developed **Flask web application**, which embeds an interactive **Tableau Public dashboard**. The application presents insights from *A College Food Choices Case Study* and has been hosted using **Render.com**, a cloud platform well-suited for deploying Python web services

11.2 Hosting Platform

- **Platform:** Render.com
- **URL:** https://render.com
- **Purpose:** To host the Flask application on a publicly accessible URL without requiring complex DevOps setup.
- **Reason for Selection:** Render provides free-tier services, native support for Python/Flask apps, easy GitHub integration, and automatic builds.

11.3 Project Structure

The Flask application was structured as follows:

```
/flask
├── app.py              # Main Flask application logic
├── requirements.txt            # Project dependencies for deployment
├── Procfile            # Specifies how to run the app using Gunicorn
├── templates/
│   └── index.html      # HTML template embedding the Tableau dashboard
├── static/             # Optional folder for CSS/JS or static assets
```

11.4 Key Configuration Files

11.4.1 requirements.txt

Defines the Python dependencies required by the project. This file ensures Render installs the correct packages during deployment.

```
Flask==2.3.2

gunicorn==21.2.0
```

11.4.2 Procfile

Instructs the Render platform to launch the Flask app using Gunicorn (a production-ready WSGI server).

```
web: gunicorn app:app
```

Note: app:app refers to the filename (app.py) and the Flask instance (app).

11.5 Deployment Process

The following steps were followed to deploy the application:

1. **Repository Setup**
2. The Flask project was uploaded to a public GitHub repository:
   📎 [github link for flaskapp]
3. **Connecting to Render**
   - Logged into Render using GitHub credentials.
   - Selected "New Web Service" and connected the repository.
4. **Configuration Settings**
   - **Build Command:** `pip install -r requirements.txt`
   - **Start Command:** `gunicorn app:app`
   - **Runtime Environment:** Python 3 (auto-detected)
5. **Automatic Build & Deployment**
   - Render cloned the repository, installed dependencies, and launched the Flask app.
   - A public URL was generated for accessing the live application.

11.6 Issue Encountered and Resolution

During the initial deployment, the following error occurred:

```
ERROR: Could not open requirements file: [Errno 2] No such file or directory: 'requirements.txt'
```

Cause: The `requirements.txt` file was missing from the repository.

Resolution:
The file was manually created with the appropriate dependencies, committed, and pushed to the GitHub repository. After re-triggering the deployment, the issue was resolved and the application deployed successfully.

11.13 Final Result

Once deployed, the Flask application successfully rendered the embedded Tableau dashboard, allowing users to interactively explore the food and nutrition data collected as part of the case study.

deployed URL: Cosmetic Insights Dashboard (wait 2 mins loading takes time)

11.8 Conclusion

The deployment process illustrates a streamlined approach to hosting data visualizations through Flask and Tableau using Render. This solution enables the delivery of dynamic dashboards to end-users via a lightweight, scalable, and cost-effective platform.

All Links

Data set : **Data Set Link**

Tableau Viz Public URL : Heritage Treasures

GitHub link for flask : github link for flaskapp

Project Demo Live link : Cosmetic Insights Dashboard (wait 2 mins loading takes time)